Data Structure and algorithms (LAB)

Lab Task

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PROBLEM 1

```
#include <iostream>

using namespace std;

void bubbleSort(int arr[], int size, char order = 'A')

{

   for (int i = 0; i < size - 1; i++)

   {

      for (int j = 0; j < size - 1 - i; j++)

      {

         if ((arr[j] > arr[j + 1] && order == 'A') || (arr[j] < arr[j + 1] && order == 'D'))

         {

            int temp = arr[j];

            arr[j] = arr[j + 1];

            arr[j + 1] = temp;

         }

      }

   }

}


void printArr(int arr[], int size)

{

   for (int i = 0; i < size; i++)

      cout << arr[i] << " ";

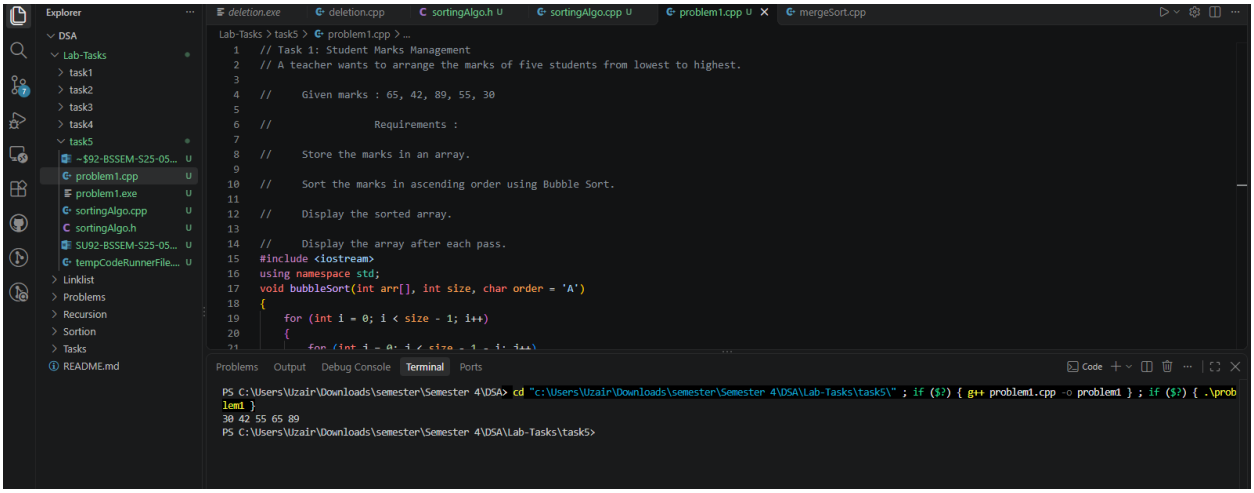
   cout << endl;

}
```

```
int main()
{
    int size = 5;
    int arr[size] = {65, 42, 89, 55, 30};
    bubbleSort(arr, size);

    printArr(arr, size);

    return 0;
}
```



PROBLEM 2

```
#include <iostream>
using namespace std;
void selectionSort(int arr[], int size, char order = 'A')
{
    order = toupper(order);
    for (int i = 0; i < size - 1; i++)
    {
        int index = i;
```

```
        for (int j = i + 1; j < size; j++)
        {

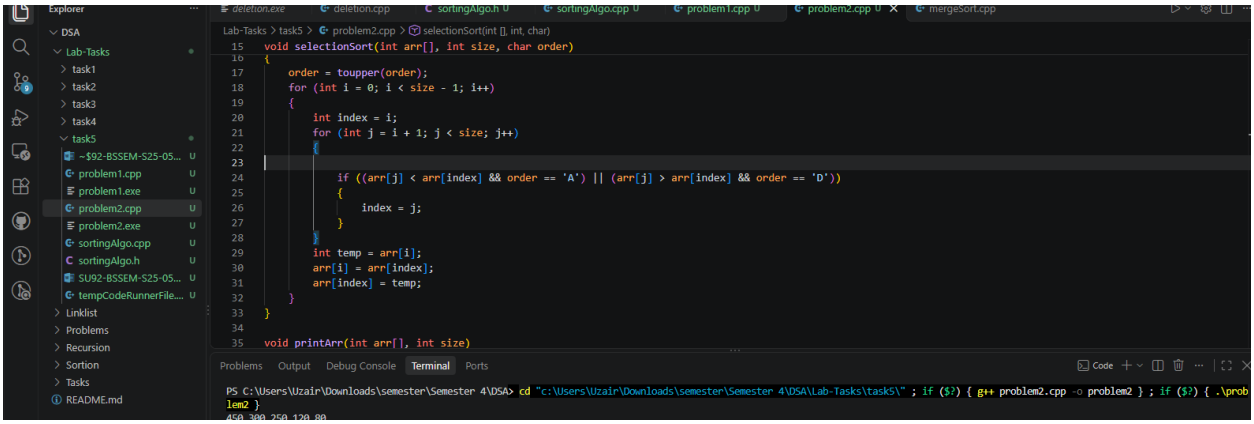
            if ((arr[j] < arr[index] && order == 'A') || (arr[j] > arr[index] && order == 'D'))
            {
                index = j;
            }
        }
        int temp = arr[i];
        arr[i] = arr[index];
        arr[index] = temp;
    }
}

void printArr(int arr[], int size)
{
    for (int i = 0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main()
{
    int size = 5;
    int arr[size] = {450, 120, 300, 80, 250};
    selectionSort(arr, size, 'D');

    printArr(arr, size);
```

```
    return 0;
}
```



PROBLEM 3

```
#include <iostream>
using namespace std;
void selectionSort(int arr[], int size, char order = 'A')
{
    order = toupper(order);
    for (int i = 0; i < size - 1; i++)
    {
        int index = i;
        for (int j = i + 1; j < size; j++)
        {

            if ((arr[j] < arr[index] && order == 'A') || (arr[j] > arr[index] && order == 'D'))
            {
                index = j;
            }
        }
        int temp = arr[i];
```

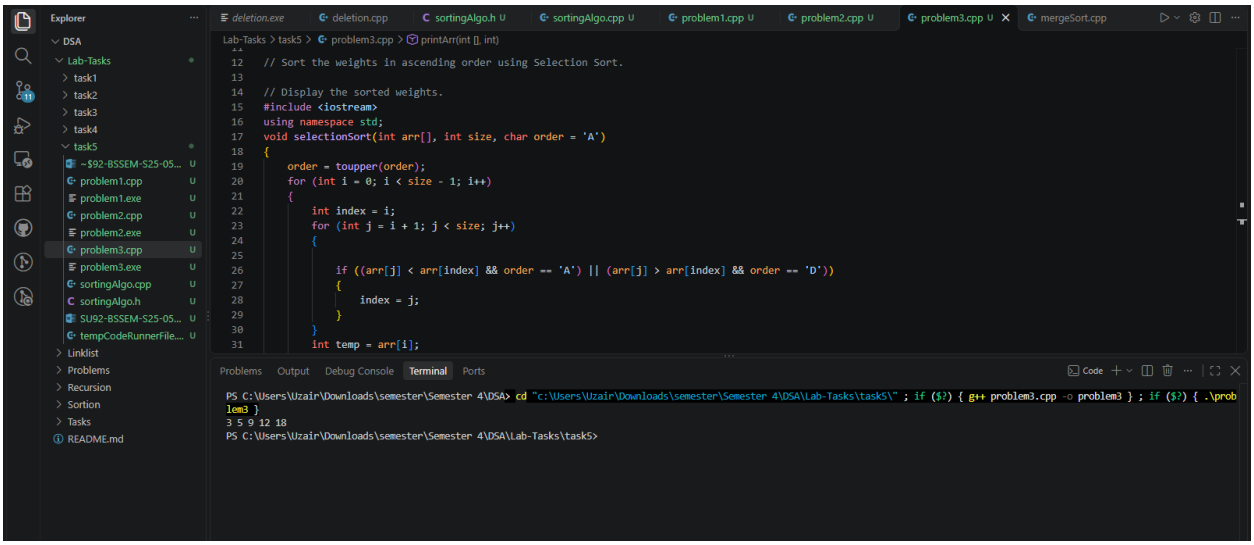
```
        arr[i] = arr[index];
        arr[index] = temp;
    }
}

void printArr(int arr[], int size)
{
    for (int i = 0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main()
{
    int size = 5;
    int arr[size] = {12, 5, 18, 3, 9};
    selectionSort(arr, size);

    printArr(arr, size);

    return 0;
}
```



PROBLEM 4

```
#include <iostream>

using namespace std;

void selectionSort(int arr[], int size, char order = 'A')

{

   order = toupper(order);

   for (int i = 0; i < size - 1; i++)

   {

      int index = i;

      for (int j = i + 1; j < size; j++)

      {


         if ((arr[j] < arr[index] && order == 'A') || (arr[j] > arr[index] && order == 'D'))

         {

            index = j;

         }

      }

      int temp = arr[i];

      arr[i] = arr[index];
```

```
        arr[index] = temp;
    }
}
void bubbleSort(int arr[], int size, char order = 'A')
{
    order = toupper(order);
    for (int i = 0; i < size - 1; i++)
    {
        for (int j = 0; j < size - 1 - i; j++)
        {
            if ((arr[j] > arr[j + 1] && order == 'A') || (arr[j] < arr[j + 1] && order == 'D'))
            {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

void printArr(int arr[], int size)
{
    for (int i = 0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main()
{
```

```
    int size = 5;
    int arr[size] = {72, 45, 90, 60, 35};
    cout << "Using Selection Sort!\n";
    selectionSort(arr, size, 'D');
    printArr(arr, size);

    cout << "Using Bubble Sort!\n";
    bubbleSort(arr, size, 'D');
    printArr(arr, size);

    return 0;
}
```

